Section 14D Comparing Data

To compare two sets of data you should:

(a) compare one of the averages (mean, mode or median) and

(b) compare the range of each set of data.

Stem and leaf plots and bar charts are used widely to show the comparison between sets of data.
A ***back to back*** stem and leaf plot compares the raw data.
Note: Always read from the stem.

Smoke		Do not smoke
	5	0 8 9
9 8 5	**6**	0 4 4 5 6 6 6 8 8
6 6 5 0 0	**7**	0 1 1 8
8 8 6 3 0	**8**	0 1 6 8 8
2 0	**9**	

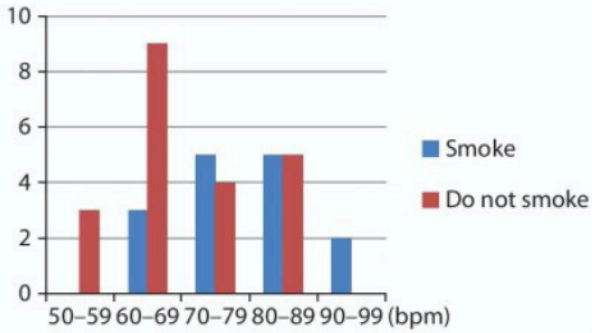
Key 8|0 = 80 bpm

read ← → read

Bar charts can be colour coded to give a quick visual comparison of two sets of data.
These two charts show the pulse rates of a number of students, taken after they had rested for 10 minutes.

A comparison was made between those who smoked and those who did not smoke

	Smoker	Non-smoker
Range	27	38
Mean	78	70



Exercise 14D

1. By calculating the (i) Range (ii) Mean, compare the following sets of data:
(a) 7, 6, 8, 4, 5, 9, 7, 3, 6, 7 (b) 6, 2, 8, 11, 3, 9, 4, 1, 5, 3

2. The back-to-back stem and leaf plot below shows the pulse rates of a class at the beginning and at the end of a P.E. class.

Pulse rates (beats per minute)

Before PE		After PE
8 7	4	
7 4 2 1 0 0	5	3 5
8 8 6 5 5 3 1	6	0 3 6 8 9
7 5 5 1 0	7	1 4 4 7 9 9
6 2 1	8	3 3 6 6 7 8 9 9
5 0	9	0 3 5 8

Key 1|8 = 81 bpm Key 6|3 = 63 bpm

(i) Write down the range of the pulses before and after the PE lesson.
(ii) Find the median of the pulses before and after the PE lesson.
(iii) Write a brief statement comparing the pulse rates of the class before and after the PE class.

3. The results of two science tests, one week apart, are shown in the following back-to-back stem and leaf plots.

Leaf	Stem	Leaf
9 8 6 6 0	3	
9 7 7 3 1 1 1	4	3 6 6 8
9 8 8 5 3 3 0	5	1 7 9 9
9 8 7 5 3	6	3 8 9
	7	0 5 5 5 8 9
	8	2 6 7 7
0	9	0 0 1 3
Key 3|6 = 63		Key 5|1 = 51

(i) What does the diagram show about the performance of the class one week later?
(ii) Find the median mark in each test.
(iii) Write down the range of the marks in each test.
(iv) Do the median marks and the range support your conclusion in (i) above?

4. By finding (a) the range (b) the median, compare the following two sets of temperatures taken on two consecutive weeks in winter.

	Week 1								Week 2						
Highest temp. (°C)	1	3	5	7	8	8	7		6	5	6	6	6	7	8
Lowest temp. (°C)	−4	−4	−2	−1	4	2	5		2	2	0	2	2	0	−1

5. The times, in minutes, taken by 15 men and 15 women to complete a crossword are shown in the following stem and leaf plots.

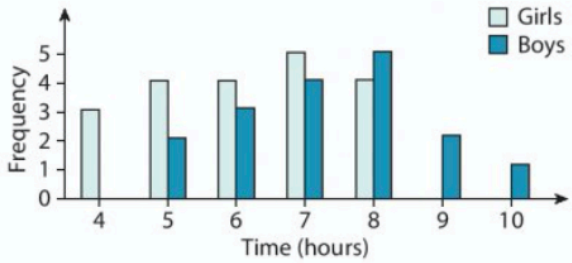
	WOMEN
1	2 3 3 5 7 7 8 9
2	4 6 8 8
3	5 6 8
4	
	Key 3|5 = 35 min

	MEN
1	2 4 4 6
2	1 3 3 3 9
3	7 8
4	3 4 5 6
	Key 2|1 = 21 min

Compare the two sets of data by finding the (a) range and (b) median of each set.
What conclusion, if any, can be made after the comparison?

6. A sample of students from 5th year were asked how many hours they spent watching television over the weekend.
The results of the survey are given in the bar chart below

Time spent watching television



(i) What was the modal time for (a) the girls (b) the boys?
(ii) What was the range in time for (a) the girls (b) the boys?
(iii) What conclusion can be drawn by comparing the two sets of data?
(iv) How many girls were included in the survey?
(v) What percentage of boys watched television for 8 hours or more?